

our search range has shrunk down to zero and our target value isn't to be found.

Notice that the code performs no actions after the recursive call returns; it immediately returns the return value of the recursive function call. This feature means that we could implement tail call optimization for this recursive algorithm, a practice we explain in Chapter 8. But also, it means that binary search can easily be implemented as an iterative algorithm that doesn't use recursive function calls. This book's downloadable resources at <https://nostarch.com/recursive-book-recursion> include the source code for an iterative binary search for you to compare with the recursive binary search.